

Instructions:

For Objective 2, your Results and Discussion should focus on **actual accomplishments completed so far**, not planned activities. Include evidence such as:

- **Software Development:** screenshots of interfaces, modules, reports, testing results, and database outputs.
- **Automation Projects:** photos of assembly, circuitry, fabrication, sensor testing, output testing, calibration, and prototype demonstrations.

Write your discussion based on what your group has already achieved and supported by the figures included in your report.

Software Development Track

Objective 2: Develop and Implement the Inventory Monitoring Module

(Insert screenshot of dashboard page)

Plate 1. Inventory Dashboard Interface

Plate 1 shows the Inventory Dashboard of the developed system. The dashboard displays the total number of products, current stock levels, low-stock alerts, and recent inventory transactions. The interface successfully retrieves data from the database and presents information in a user-friendly format. Initial testing confirmed that inventory updates are reflected in the dashboard in real time.

(Insert screenshot of Add Product page/form)

Plate 2. Inventory Entry Module

Plate 2 presents the Inventory Entry Module where administrators can register new products. Required fields include product name, category, quantity, and unit price. Validation mechanisms were implemented to prevent incomplete data submission. Testing showed that records are successfully stored in the database upon submission.

(Insert screenshot of product list table)

Plate 3. Product Management Interface

Plate 3 illustrates the Product Management Interface. This module allows users to view, edit, and delete inventory records. Functional testing revealed that modifications made to product information are accurately updated within the database and reflected in the system interface.

(Insert screenshot showing search results)

Plate 4. Search and Filter Function

Plate 4 shows the implemented search functionality. Users can quickly locate inventory records using keywords. Preliminary testing demonstrated that the system correctly filters matching records and improves data retrieval efficiency.

(Insert screenshot of successful test cases or transaction logs)

Plate 5. System Testing Results

Plate 5 presents sample testing results of the inventory module. All tested functions, including product addition, editing, deletion, and searching, produced expected outputs. These results indicate the successful implementation of the current module.

Automation / Hardware-Based Track

Objective 2: Develop and Integrate the Automated Irrigation System

(Insert photo of assembled prototype)

Plate 1. Prototype Assembly

Plate 1 shows the initial assembled prototype of the Automated Irrigation System. The system consists of a microcontroller, soil moisture sensor, relay module, water pump, and power source. All components were securely connected according to the designed circuit configuration.

(Insert photo of wiring and connections)

Plate 2. Circuit Wiring Configuration

Plate 2 presents the completed wiring configuration of the prototype. Proper connections among the input, processing, and output components were verified through continuity testing. The wiring setup provided stable communication among the system components.

(Insert photo/screenshot of sensor readings)

Plate 3. Soil Moisture Sensor Testing

Plate 3 demonstrates the testing of the soil moisture sensor. The sensor generated different readings when exposed to dry and moist soil conditions. The obtained values confirmed the sensor's capability to detect variations in soil moisture levels.

(Insert photo showing activated pump or relay indicator)

Plate 4. Relay and Water Pump Activation Test

Plate 4 illustrates the activation test of the relay-controlled water pump. Results indicated that the relay was triggered whenever the measured soil moisture level fell below the programmed threshold. The water pump automatically supplied water until the desired moisture level was achieved.

(Insert screenshot of Arduino IDE Serial Monitor or equivalent)

Plate 5. Serial Monitor Output

Plate 5 presents the serial monitor output during system testing. The displayed data include sensor readings and corresponding system actions. The recorded outputs verify successful communication between the sensor and microcontroller.

(Insert photo of actual deployment/testing environment)

Plate 6. Initial Field Testing

Plate 6 shows the prototype during initial field testing. The prototype responded appropriately to changing soil conditions and executed irrigation actions according to programmed parameters. Further testing will be conducted to evaluate long-term reliability and efficiency.

Submission Requirement

Submit an updated Chapter 3 containing:

1. Completed Results and Discussion for Objective 1.
2. Initial Results and Discussion for Objective 2.
3. Supporting evidence using properly labeled plates.

Software Development Projects:

- At least **5 plates** of system development progress (interfaces, modules, reports, testing results).

Automation/Hardware Projects:

- At least **6 plates** showing prototype assembly, circuitry, component testing, integration, and initial deployment/testing.

ALL SUBMISSIONS MUST BE UPLOADED TO YOUR INDIVIDUAL FOLDERS PROVIDED EARLIER ON GOOGLE DRIVE UNTIL MONDAY, OCTOBER 5, 2026 READY FOR PER TEAM CHECKING DURING OUR MUTUAL VACANT HOURS. PLEASE OBSERVE PROPER TIME MANAGEMENT.